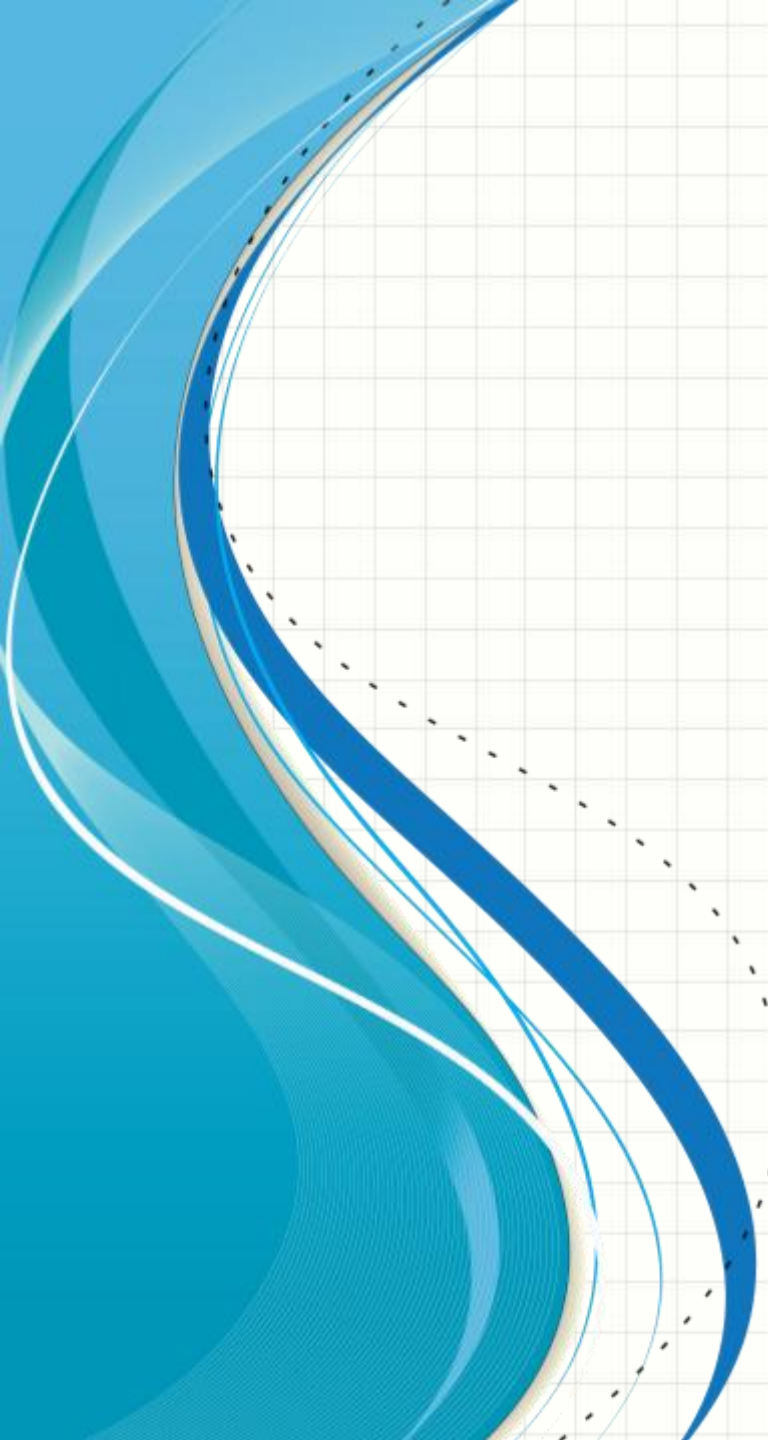




# **CHAPTER # 5**

## **SEQUENTIAL LOGIC**

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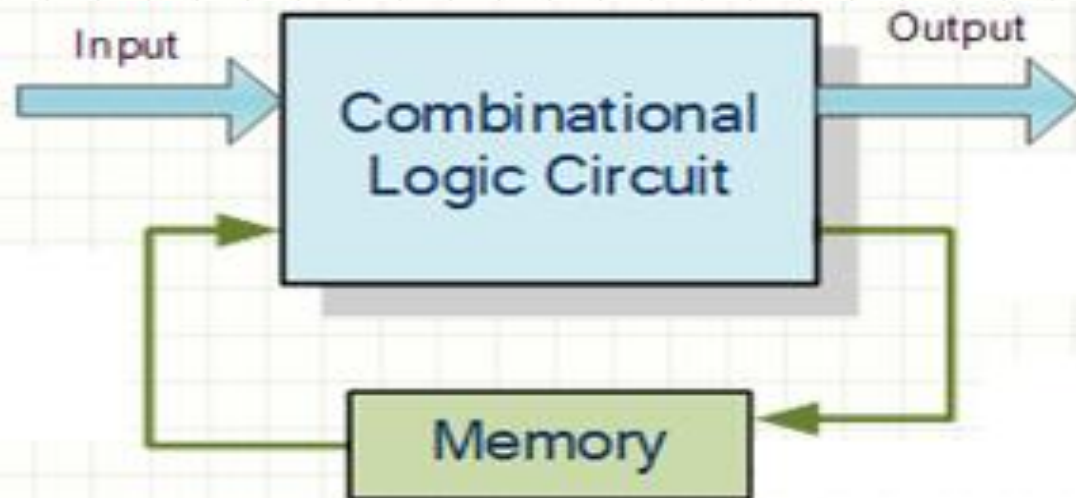
# Objective of Lecture

After studying this chapter, you should be able to know

- Introduction.
- Sequential logic circuits & types.
- The Clock.
- Flip-Flop.
- Triggering Of Flip-Flops.
- Types of Flip-Flop.

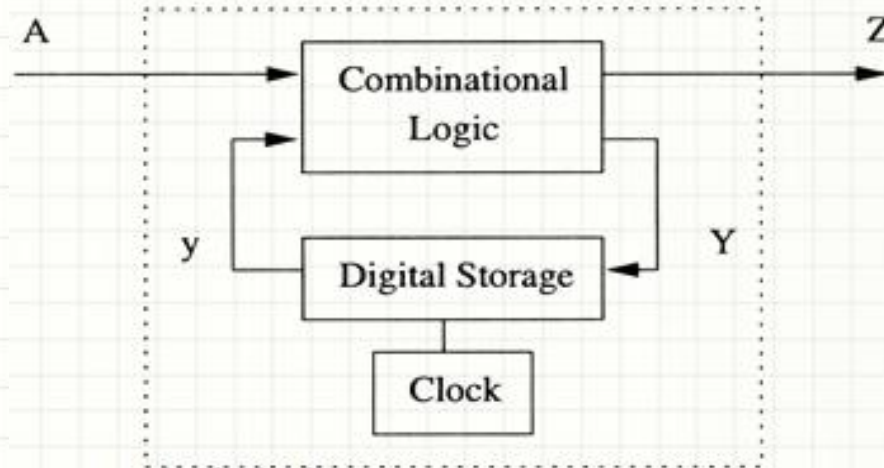
# Introduction Sequential circuits

- Consists of logic gates + storage elements.
- Output = Present inputs and previous outputs.
- Input must follow specific sequence to produce output.
- In order to follow sequence of input circuit must contain memory to retain knowledge of those inputs.
- Memory is obtained by feedback connection.
- In a sequential circuit data is stored in memory devices called flip-flops.

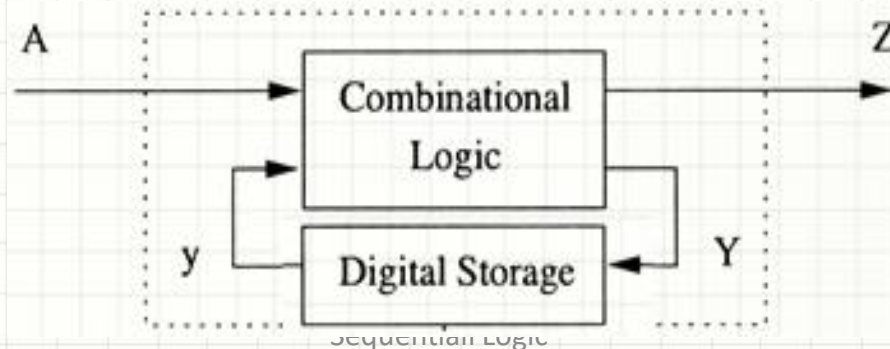


# Sequential Logic Circuits

- **Two Types of sequential circuit**
- Synchronous types (level or clock pulse to control the storage element).

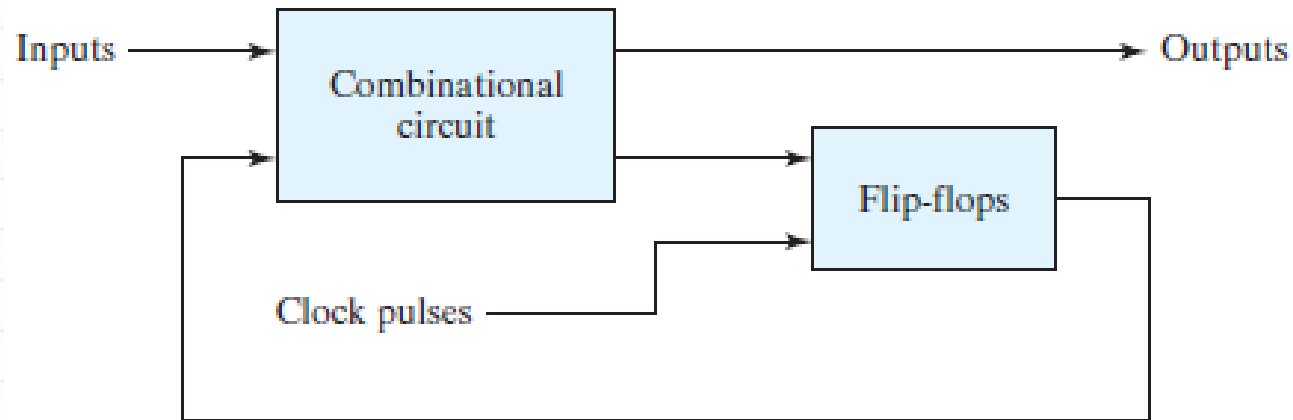


- Asynchronous circuit (uses instructions/signals that indicate completion of operation or protocols)



# Synchronous Sequential circuit

- Synchronous sequential circuits that use clock pulses to control storage elements are called Synchronous clocked sequential circuits, because the activity within the circuit and the resulting updating of stored values is synchronized to clock pulse.



(a) Block diagram



(b) Timing diagram of clock pulses

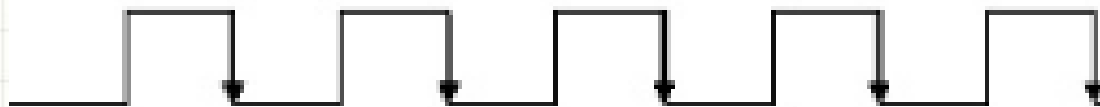
Synchronous clocked sequential circuit

# The CLOCK

- In synchronous device the exact time at which output change its state are controlled by a signal called clock (clk).
- A clock signal is generally rectangular pulse train or square wave.
- Output change state when clock make a transition.
- LOW state to HIGH state is called positive going transition (PGT).
- HIGH state to LOW state is called negative going transition (NGT).



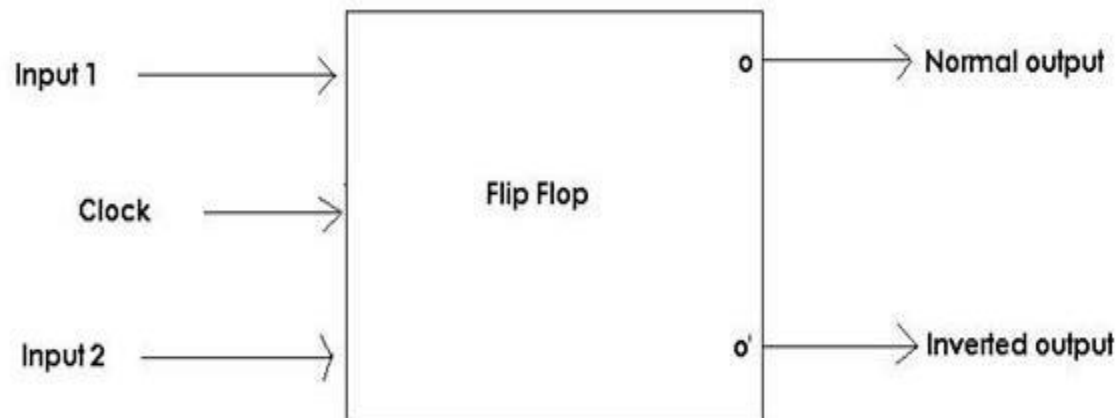
**(a) Positive going transition**



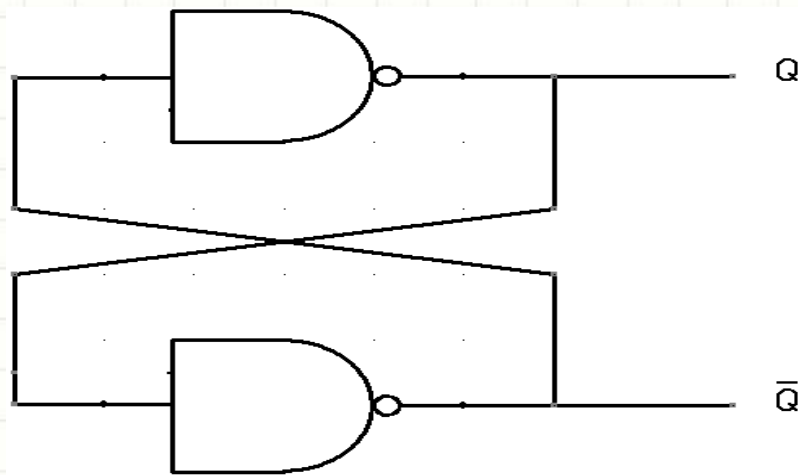
**(b) Negative going transition**

# Flip-Flop

- A binary storage device capable of storing one bit of information either logic 0 or logic 1.
- A flip-flop has two stable states and hence it is also called as bistable multi-vibrator.
- We can construct it using both NAND or NOR gates.
- A flip-flop circuit normally consists of :



# Basic Flip-Flop



- **Conclusions:**
- The two outputs are complement of each other.
- It has two stable states 0 state(reset) and 1 state (set).
- If the circuit is in 0 state, it continues to remain in this state. Similarly if it is in 1 state, it continues to remain in this state. This property of the circuit is called memory.
  - There is no way to enter desire information, so we use two I/P gate.
  - When we power ON the circuit it switches to one of the stable state.

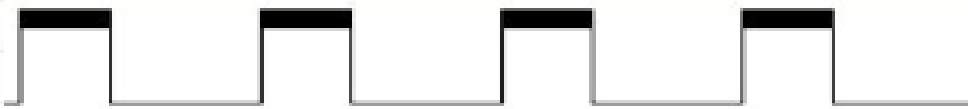
# Triggering Of Flip-Flops

- We already know that a flip-flop has two stable states.
- A flip-flop has 0 as its initial value and goes momentarily to 1 and returns back to 0.
- The state of the flip-flop can be changed by applying a momentary change to the Input signal. This momentary change is called triggering of a flip-flop.
- Flip-Flops are triggered by the clock pulse. A clock pulse is usually a square wave.
- Triggering of flip-flops can be classified into:
  1. Level Triggering
  2. Edge Triggering

# Level Triggering

## ➤ Level Triggering:

- Level triggered flip-flops are dependent on the period of the pulse applied to it. It is further classified into two types.
  - Positive Level Triggering



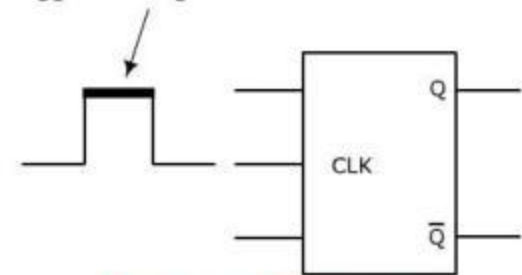
Response to positive level

## – Negative Level Triggering



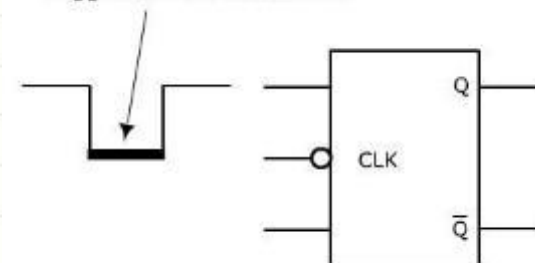
Response to Negative level

Triggers on high clock level



**High Level Triggering**

Triggers on low clock level



**Low Level Triggering**

# Edge Triggering

## ➤ Edge Triggering

- The edge triggered flip flops changes state either at positive edges or negative edges of the clock pulse applied.
- They are classified into:
  - Positive edge triggered Flip flops



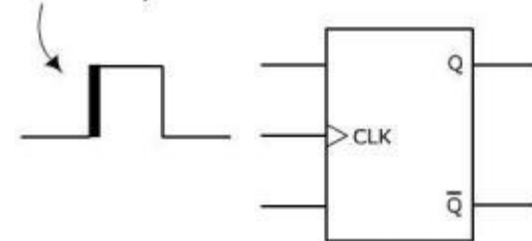
Positive-edge response

- Negative edge triggered flip flops



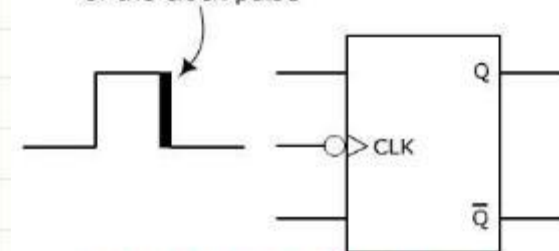
Negative-edge response

Triggers on this edge of the clock pulse



**Positive Edge Triggering**

Triggers on this edge of the clock pulse



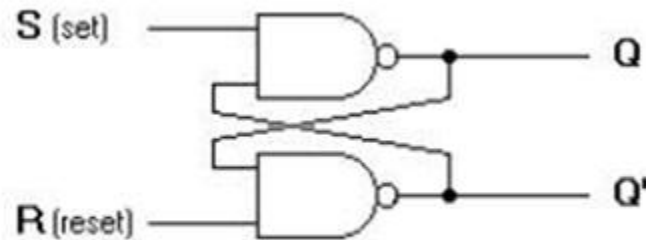
**Negative Edge Triggering**

# Types of Flip-Flop

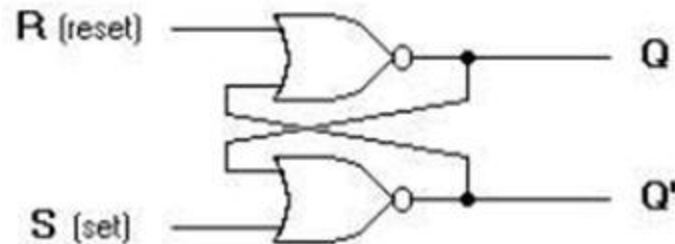
- They can be classified according to the number of inputs they possess and the manner in which they affect the binary state of the flip-flop.
- SR flip-flop
- D flip-flop
- JK flip-flop
- T flip-flop

# Basic Flip-Flop

Can be constructed with the help of two NAND or NOR gates.

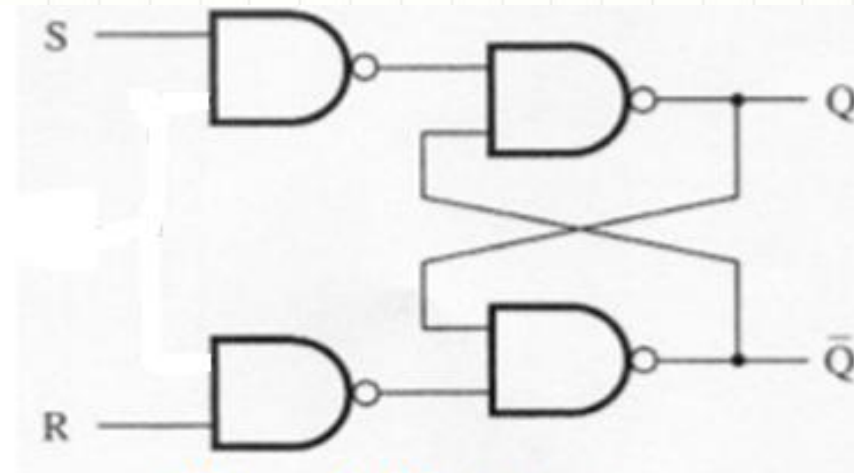


Basic flip-flop circuit with NAND gates



Basic flip-flop circuit with NOR gates

# External Input SR Flip-Flop



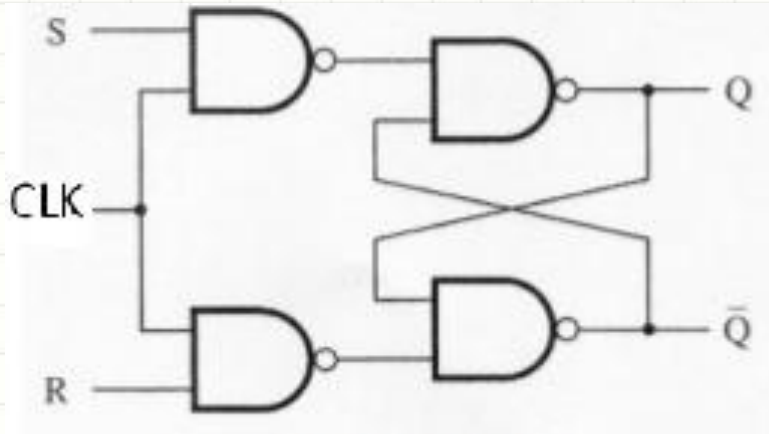
**Circuit Diagram**

| Inputs |   | Outputs   |                 | Comments      |
|--------|---|-----------|-----------------|---------------|
| S      | R | $Q_{n+1}$ | $\bar{Q}_{n+1}$ |               |
| 0      | 0 | $Q_n$     | $\bar{Q}_n$     | No change     |
| 0      | 1 | 0         | 1               | Rset          |
| 1      | 0 | 1         | 0               | Set           |
| 1      | 1 | x         | x               | Indeterminate |

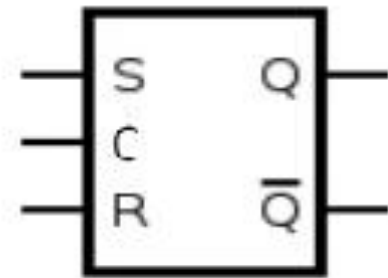
This type of flip-flop is called SR flip flop or RS flip flop.

# Clocked SR Flip-Flop

It is often required to set the memory cell in synchronous with clock such circuit is called clocked SR flip flop.



**Circuit Diagram**

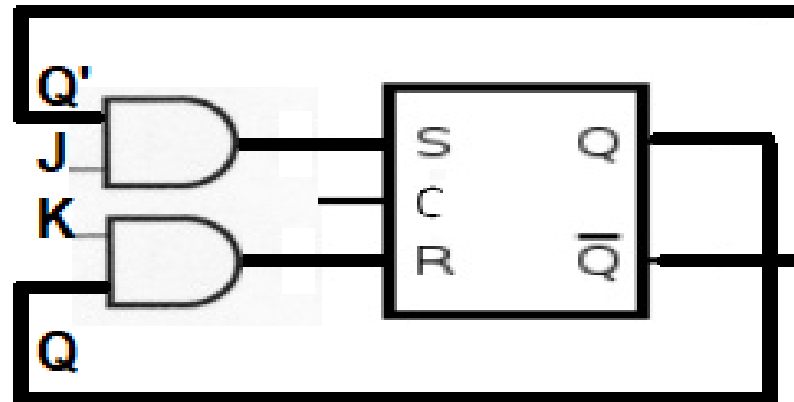


**Block Diagram**

| Inputs |   |   | Outputs           |                         | Comments       |
|--------|---|---|-------------------|-------------------------|----------------|
| clk    | S | R | Q                 | $\bar{Q}$               |                |
| 0      | x | x | $Q_{\text{prev}}$ | $\bar{Q}_{\text{prev}}$ | Previous Value |
| 1      | 0 | 1 | 0                 | 1                       | Rset           |
| 1      | 1 | 0 | 1                 | 0                       | Set            |
| 1      | x | x | Q                 | Q'                      | Previous Value |
| 1      | 1 | 1 | x                 | x                       | Indeterminate  |

# JK Flip-Flop

- In SR flip-flop we have invalid value for  $S=R=1$ . So we made modification to overcome this issue



**Block Diagram**

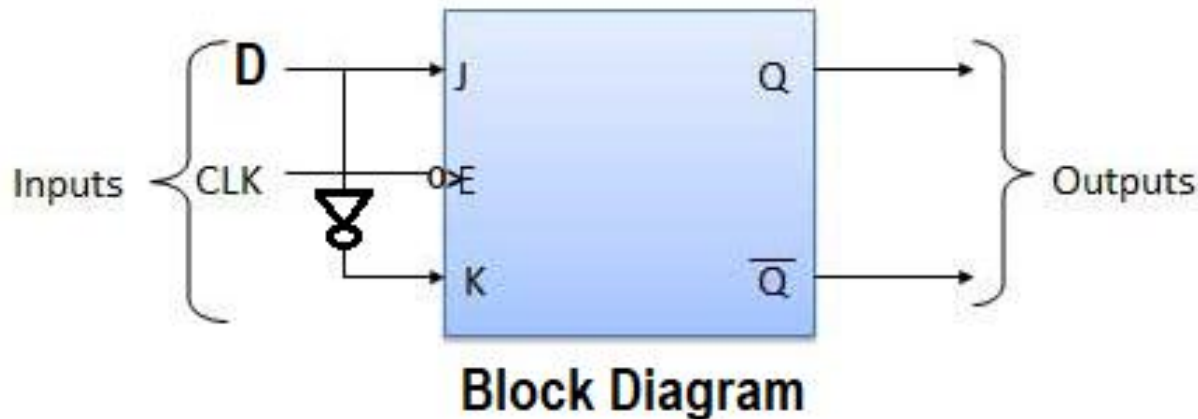
| J | K | Q | $\overline{Q}$ | $\overline{\text{SET}}$ | $\overline{\text{RESET}}$ | Q | $\overline{Q}$ |
|---|---|---|----------------|-------------------------|---------------------------|---|----------------|
| 1 | 0 | 1 | 0              | 0                       | 0                         | 1 | 0              |
| - | - | 0 | 1              | 1                       | 0                         | 1 | 0              |
| 0 | 1 | 1 | 0              | 0                       | 1                         | 0 | 1              |
| - | - | 0 | 1              | 0                       | 0                         | 0 | 1              |
| 0 | 0 | 1 | 0              | 0                       | 0                         | 1 | 0              |
| - | - | 0 | 1              | 0                       | 0                         | 0 | 1              |
| 1 | 1 | 1 | 0              | 0                       | 1                         | 0 | 1              |
| - | - | 0 | 1              | 1                       | 0                         | 1 | 0              |

# D Flip-Flop

- D flip flop can be constructed by both SR and JK flip flop.
- It works as a memory cell.

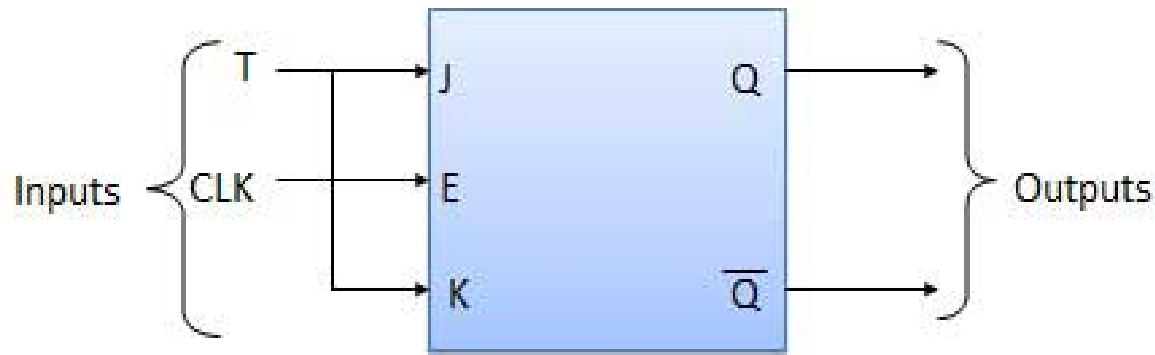
$$I/P = O/P$$

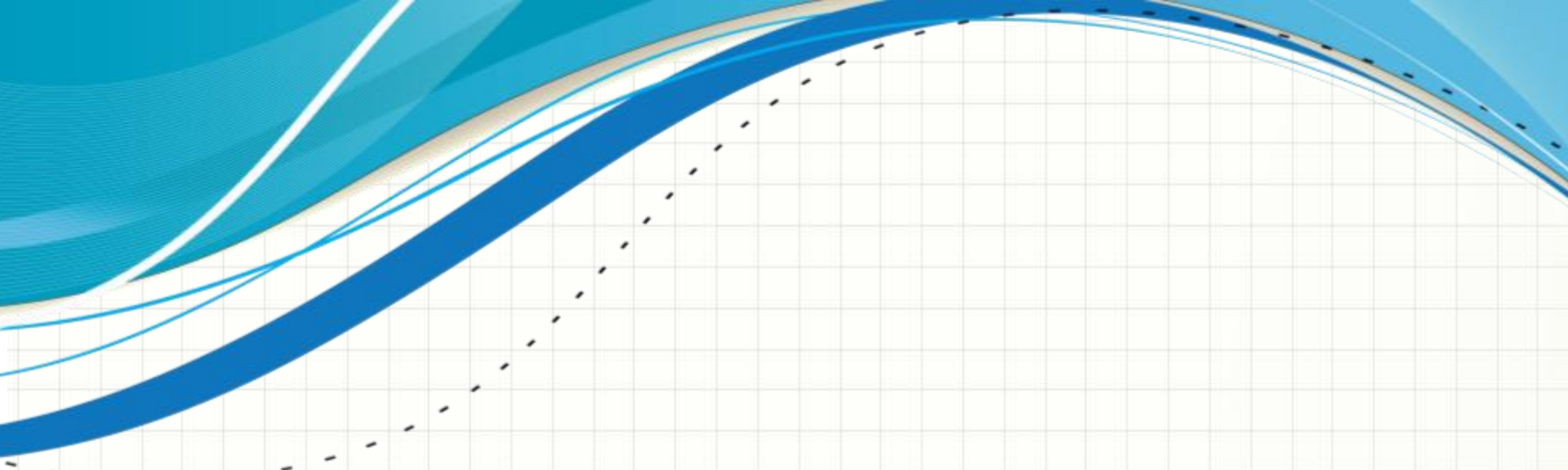
Delay (Negative edge trigger)



# T Flip-Flop

- It gives the toggle value when input=1.





# THANKS